

Exploring the role of microRNAs in axolotl regeneration

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Abstract

The axolotl, *Ambystoma mexicanum*, is used extensively for research in developmental biology, particularly for its ability to regenerate and restore lost organs, including in the nervous system, to full functionality. Regeneration in mammals typically depends on the healing process and scar formation with limited replacement of lost tissue. Other organisms, such as spiny mice (*Acomys cahirinus*), salamanders, and zebrafish, are able to regenerate some damaged body components. Blastema is a tissue that is formed after tissue injury in such organisms and is composed of progenitor cells or dedifferentiated cells that differentiate into various cell types during regeneration. Thus, identifying the molecules responsible for initiation of blastema formation is an important aspect for understanding regeneration. Introns, a major source of noncoding RNAs (ncRNAs), have characteristic sizes in the axolotl, particularly in genes associated with development. These ncRNAs, particularly microRNAs (miRNAs), exhibit dynamic regulation during regeneration. These miRNAs play an essential role in timing and control of gene expression to order and organize processes necessary for blastema creation. Master keys or molecules that underlie the remarkable regenerative abilities of the axolotl remain to be fully explored and exploited. Further and ongoing research on regeneration promises new knowledge that may allow improved repair and renewal of human tissues.

KEYWORDS

axolotl, microRNA, regeneration, salamander, transcriptome

1 | INTRODUCTION

The axolotl (*Ambystoma mexicanum*) is an important model for studying regeneration. The animal displays unique regenerative abilities. It can regenerate many body tissues, including vital organs, and can restore generated organs to full functionality. The axolotl genome contains many repeated elements that increase the size of introns and intergenic regions (10× longer than orthologous vertebrate introns) (Nowoshilow et al., 2018; J. J. Smith et al., 2009). Furthermore, introns of developmental genes in the axolotl are shorter than those of nondevelopmental genes compared with introns in the human, mouse, and frog genome. In addition, genes that

contain many exons and only short introns show Gene Ontology enrichment related to developmental signaling that is not enriched in similar genes with larger introns, suggesting that the intron size of developmental genes in the axolotl is constrained (Nowoshilow et al., 2018). Introns are suggested to be a major source of noncoding RNAs (ncRNAs; J. J. Smith et al., 2009; St Laurent et al., 2012). The intronic ncRNAs are denser in axolotls than in humans (J. J. Smith et al., 2009). Thus, ncRNAs could be key mediators in the regeneration process.

MicroRNAs (miRNAs) are small ncRNAs, ~22 nucleotides in length, that regulate the expression of many genes and mediate responses to diseases, including cancer. In addition, miRNA may be

Abbreviations: BMP4, bone morphogenetic protein 4; CRABP, cellular retinoic acid binding protein; miRNAs, microRNAs; MMPs, matrix metalloproteinases; ncRNAs, noncoding RNAs; NF-κB, nuclear factor kappa light-chain enhancer of activated B cells; RA, retinoic acid; ROS, reactive oxygen species; Smo, smoothened; UTR, untranslated region.

involved in the behavior and health status of the central nervous system (Abo-Al-El, 2019b; M. Li et al., 2020; J. Li et al., 2020). miRNAs can alter the basal state of cells, for example, by regulating the maturation, proliferation, and differentiation of cells responsible for immune responses. Such activities involve interaction with other regulatory molecules, such as transcription factors (Lu & Liston, 2009). Finally, miRNA may play a central regulatory and effector role in reprogramming cells to initiate an early regenerative response (Sehm, Sachse, Frenzel, & Echeverri, 2009). Such miRNAs are selectively summarized in Table 1. miRNAs can directly or indirectly orchestrate the regeneration processes through several related signaling pathways, such as cytokine and hedgehog signaling (Singh

et al., 2015). Details of these regeneration-related pathways are difficult to characterize.

miRNAs are a new class of conventional, rare, genomic regulatory molecules, and the most conserved genetic elements in the genome (Sempere, Cole, Mcpeek, & Peterson, 2006) could become phylogenetic markers for studying evolutionary relationships (Tarver et al., 2013). The de novo origination of ncRNA such as miRNAs can occur at a greater rate than that in protein coding genes (Wu & Zhang, 2013). Furthermore, mature miRNA genes evolve twice more slowly than most conserved positions of protein coding genes (Tarver et al., 2013). Several key regulatory miRNAs, including those related to regeneration, are conserved between axolotl and mammals (Diaz

TABLE 1 MicroRNAs associated with regenerative process in different species

MicroRNA	Direction of expression	Organ and species	Reference
miR-200a	Upregulation	Spinal cord, axolotl	Sabin et al. (2019)
mir-122, miR-233, miR425-5p, and miR3600	Upregulation	Liver, axolotl	Caballero-Pérez et al. (2018)
miR-429 and miR-454b	Downregulation		
miR-206	Downregulation	Tail, axolotl	Gearhart et al. (2015)
let-7a, Amex-miR-pn2533, -pn3918, and -pn2611	Upregulation	Tail, axolotl	Gearhart et al. (2015)
miR-196	Upregulation	Tail, axolotl	Sehm et al. (2009)
miR-21	Upregulation	Tail and limb, axolotl	Gearhart et al. (2015); Holman et al. (2012)
		Caudal fin, bichir pectoral fin, zebrafish	King and Yin (2016)
miR-125b	Upregulation	Spinal cord, axolotl	Diaz Quiroz et al. (2014)
		Brainstem, zebrafish	Y. M. Yu et al. (2011)
let-7i	Upregulation	Heart, zebrafish	A. M. Smith et al. (2019)
let-7	Downregulation	Retina, zebrafish	Kaur et al. (2018)
miR-26a	Downregulation	Heart, zebrafish	Crippa et al. (2016)
miR-101a	Downregulation	Heart, zebrafish	Beauchemin, Smith, and Yin (2015)
let-7a/c and miR-99/100	Downregulation	Heart, zebrafish	Aguirre et al. (2014)
miR-133	Downregulation	Heart and fin, zebrafish	Yin et al. (2012); Yin et al. (2008)
miR-133b	Upregulation	Spinal cord, zebrafish	Y. M. Yu et al. (2011)
		Tail, green anole lizard	Hutchins et al. (2016)
miR-133a	Upregulation	Tail, green anole lizard	Hutchins et al. (2016)
miR-21 and miR-142	Upregulation	Limb, <i>Cynops orientalis</i>	Y. Yu et al. (2019)
miR-223 and miR-133a	Downregulation	Limb, <i>Cynops orientalis</i>	Y. Yu et al. (2019)
miR-9 cluster	Upregulation	Oral siphon, <i>Ciona robusta</i>	Spina, Guzman, Zhou, Kosik, and Smith (2017)
miR-10b-5p	Downregulation	Limb, newt	Brito (2018)
miR-128	Downregulation	Heart, mice	Huang et al. (2018)
miR-138-5p	Upregulation	Peripheral nerve, rat	Sullivan et al. (2018)
miR-127	Downregulation	Liver, rat	Pan et al. (2012)

Quiroz, Tsai, Coyle, Sehm, & Echeverri, 2014). For example, most miRNA precursors expressed during tail regeneration in the green anole lizard (*Anolis carolinensis*) are conserved and likely have homologs in other vertebrates (Hutchins, Eckalbar, Wolter, Mangone, & Kusumi, 2016). Given these observations, miRNAs could be a key to understanding regeneration in vertebrates.

Key molecules involved in axolotl regeneration remain unknown. Regeneration is a complex process that involves many factors (Figure 1), including genes that may be involved in regeneration, such as *Prod1* (Garza-Garcia, Driscoll, & Brockes, 2010) and fibroblast growth factor ligand gene, *fgf20a* (Whitehead, Makino, Lien, & Keating, 2005; Yin et al., 2008); stem and progenitor cells; dedifferentiation, transdifferentiation, and reprogramming of adult cells (Poss, 2010); transcription factors; ncRNAs; and epigenetics (Cieřlar-Pobuda et al., 2017). Dedifferentiation of cells after injury is often proposed as the process for blastema formation (Singh et al., 2015). Interaction between epigenetics and miRNAs has been reported previously (Pal, Ghatak, & Sen, 2015). Furthermore, miRNAs and transcription factors could be important for transdifferentiation, in which miRNAs alone or in combination with other transcription factors reprogram differentiated cells to pluripotent cells (Figure 2; Sen & Ghatak, 2015).

More specifically, axolotls grafted with skeletal muscle, cartilage tissues, dermis cells, or Schwann cells from transgenic axolotls that express green fluorescent protein, GFP, formed their tissue of origin during regeneration. Tissue formation was confirmed using tissue-specific methods such as PAX7⁺ and Myf5⁺ labeling (Kragl et al., 2009). Kragl et al. (2009) concluded that the blastema is composed of heterogeneous restricted progenitor cells and that limb regeneration is accomplished with partial dedifferentiation without giving rise to pluripotent cells. Each mature cell retains its cell-type

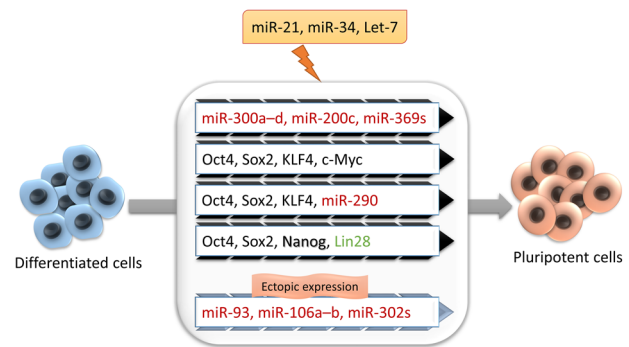
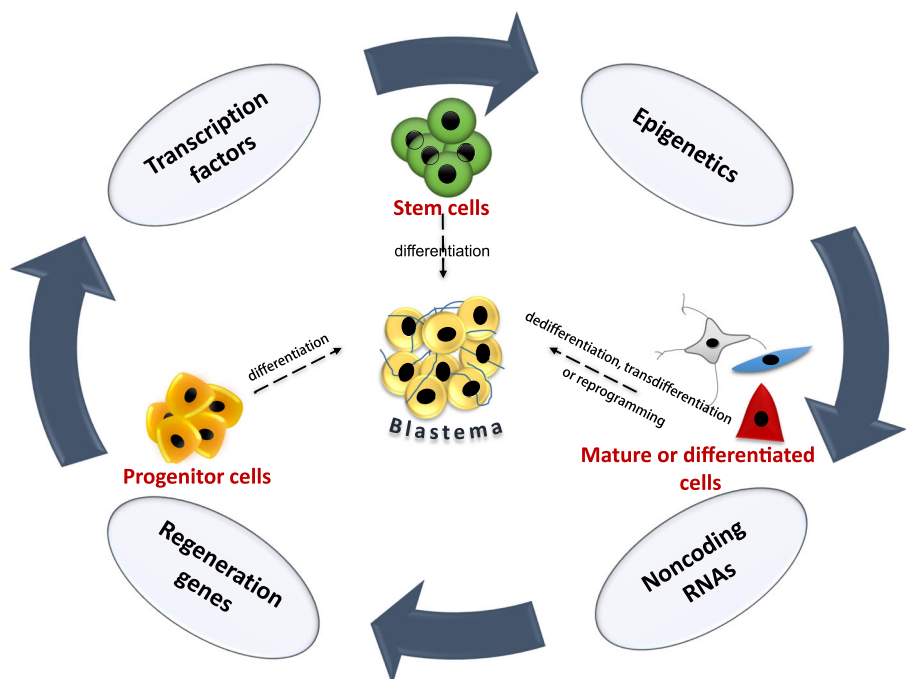


FIGURE 2 Role of microRNAs and transcription factors and other genes in induction of pluripotency. Expression of miR-300a-d, miR-200c, and miR-369s, the transcription factors “Yamanaka factors” Oct4, Sox2, KLF4, and c-Myc, and Nanog, the RNA-binding protein Lin28, and ectopic expression of miR-93, miR-106a-b, and miR-302s, in combinations as illustrated, can induce pluripotency in differentiated cells. Overexpression of miR-290 could replace c-Myc in transcription factor combinations. Lin28 binds to the let-7 pro- and pre-miRNA, which blocks the production of the mature let-7. Expression of miR-21, miR-34, let-7 inhibits transition to pluripotency (Sen & Ghatak, 2015)

identity during axolotl limb regeneration without transforming to pluripotency.

Thus, cells are induced toward pluripotency without losing their epigenetic imprinting in a manner similar to in vitro induction to pluripotency of adult cells. In addition, connective tissue cells were found to organize in regenerated tissue using positional memory (Bryant, Endo, & Gardiner, 2002; McCusker & Gardiner, 2014). During axolotl limb regeneration, cartilage cells have proximo-distal positional identity but Schwann cells do not (Kragl et al., 2009).

FIGURE 1 Schematic and hypothetical illustration depicting potential modulators of the formation of blastema, and regeneration in axolotl. All or some of these factors are implicated in the regeneration process. Interaction and cross signaling among these factors have been evidenced. Interrupted lines refer to possible contributions, and other factors, such as noncoding RNAs refer to proven modulators



Positional memory is not well understood and could be a key to understanding regeneration. Rinn, Bondre, Gladstone, Brown, and Chang (2006) suggested a model to explain how cells retain positional knowledge. This "ZIP-code system" proposes that groups of cells in a given location utilize a set of genes (acts as a code) unique to that position and related to anatomic axes. Interestingly, fibroblasts, a primary part of healing processes, retain an important feature of the embryonic code, *HOX* expression, and retain epigenetic information responsible for anatomically specific differentiation (Rinn et al., 2006). Connective tissue cells, likely fibroblasts, have been shown to dedifferentiate into common, multipotent progenitors with an embryonic limb bud cell-like phenotype upon the induction of limb regeneration in axolotls, which will later redifferentiate into diverse cell types (Gerber et al., 2018; Leigh et al., 2018). Fibroblasts are critical for positional identity. The fibroblast growth factor receptor controls miRNA expression during zebrafish fin regeneration (Yin et al., 2008).

In contrast, known dedifferentiation marker genes—*msx1*, *Nrad*, *Sox2*, *Klf4*, *c-Myc*, and *LIN28*—were not targets of differentially expressed miRNAs during limb regeneration in the salamander, *Cynops orientalis* (Y. Yu et al., 2019). Apparently, each cell retains its original features and is responsible for regeneration of specific tissue types. Nevertheless, contributions of dedifferentiated and transdifferentiated cells cannot be excluded.

Specific cell types and effectors that initiate the regeneration process are thus not clear in the available literature. This review discusses dynamic miRNA regulation of regeneration processes with emphasis on the axolotl. Figure 3 compiles candidate miRNAs that have been studied during axolotl regeneration. Studies on other species are included to add layers of understanding to the function of miRNAs.

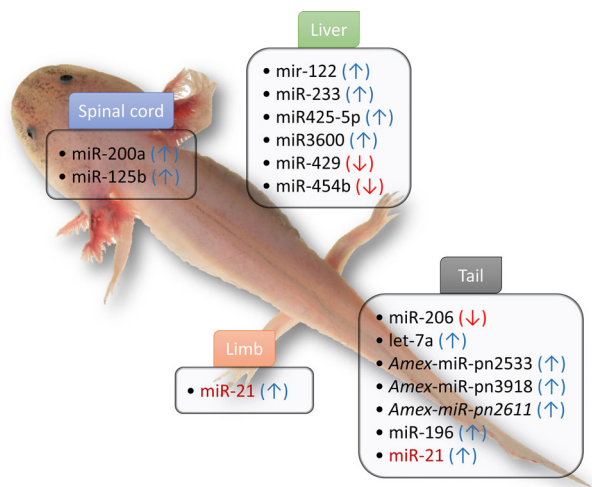


FIGURE 3 The different miRNAs that have been discovered during axolotl regeneration. Candidate miRNAs that have been studied when spinal cord, tail, limb regenerate show that some are upregulated and others downregulated. Of these, miR-21 showed an upregulation during both tail and limb regeneration. miRNAs, micro RNAs

2 | AXOLOTL SPINAL CORD REGENERATION

miRNAs expressed during regeneration are mostly shared among tissues, although brain tissues showed the largest number of unique miRNAs in the anole lizard (Hutchins et al., 2016). The roles of miRNAs in axon regeneration in diverse species has been reviewed by Li, Teng, and Liu (2016) and Quinzaños-Fresnedo and Sahagún-Olmos (2015). However, a similar role in the regeneration of axolotl spinal cord is not completely understood. Transplantation of GFP⁺ spinal cord into non-GFP axolotl showed that a region within 500 μm of an amputation plane is the source of neural progenitors for regeneration (Mchedlishvili, Epperlein, Telzerow, & Tanaka, 2007). The size of this zone does not depend on the amputation plane even if the plane is along the anteroposterior axis (Tazaki, Tanaka, & Fei, 2017). Following injury, GFAP⁺ glial cells in spinal cords of the axolotl start to proliferate and migrate to form a neural tube that initiates axon regeneration. A key factor in this process is miR-200a, which plays a unique role in differential regulation of genes. For example, miR-200a represses *c-Jun* expression by targeting the 3'-untranslated region (3'-UTR) and thereby blocks the formation of AP-1^{cFos/cJun}. This latter factor is involved in glial scar formation, reactive gliosis, extracellular matrix remodeling, and axon guidance. Upregulation of *c-Fos* in glial cells and the ability of *c-Fos* to homodimerize to form AP-1 is essential for the proliferation of glial cells. The axolotl expresses a specific JUN protein, JunB, that heterodimerizes with *c-Fos* in axolotl glial cells. Thus, unlike mammals, the axolotl expresses a noncanonical AP-1^{cFos/JunB} after spinal cord injury (Sabin, Jiang, Gearhart, Stewart, & Echeverri, 2019), and this pathway depends on miRNA.

In a comparative study with the axolotl and rat, Diaz Quiroz et al. (2014) found that precise control of miR-125b expression is essential for proper regeneration of the spinal cord, especially reconnection of axons through direct targeting of *Sema4D*. Interestingly and consistent with this finding, induction of miR-125b downregulated the expression of *Sema4D* and other glial scar-related genes, including *GFAP*, *CSPG4*, and *Col6A1*, and enhanced functional recovery in rats. However, miR-125b repressed scar-related genes, a function contrary to healing in mammals, which depends on scar formation (Diaz Quiroz et al., 2014).

In contrast, experimental reduction of miR-125b expression in the axolotl to levels found in the rat resulted in failure of axons to cross the injury site, indicating a significant role played by miR-125b in spinal cord regeneration (Diaz Quiroz et al., 2014). Similarly, miR-133b plays an important role in spinal cord regeneration in zebrafish and green anole lizard (Hutchins et al., 2016; Y. M. Yu et al., 2011). miR-133b is upregulated in regenerating neurons of the brainstem and promotes tissue regeneration in other tissues. Inhibition of miR-133b expression reduces regeneration of axons from neurons in superior reticular and intermediate reticular formations and in the nucleus of the medial longitudinal fascicle, which results in a decrease in the number of neurons with regenerated axons in the brainstem. miR-133b targets the RhoA protein, an inhibitor of axonal growth,

and neurite outgrowth-related molecules. Strong expression of miR-133b is recorded in motor neurons; hence, its inhibition impairs locomotor recovery after injury (Y. M. Yu et al., 2011).

2.1 | Different targets and roles of miRNAs in different contexts and even in different locations within the same tissue

miRNAs levels may show differences among locations within the same organism during regeneration. Eleven miRNAs, including miR-133a, miR-133b, and miR-206, were differentially expressed between the regenerating tail tip and base during maximum outgrowth. Levels of miR-133a were higher in anole regenerating tail base than in the tail tip (Hutchins et al., 2016). Differentially expressed messenger RNAs (mRNAs) could reflect the presence of different tissues in different portions of the tail. For example, skeletal muscle is prevalent in the tail base, in which growth and differentiation-related miRNAs are highly expressed (Hutchins et al., 2016).

In zebrafish, downregulation of miR-133 is essential for the regeneration of heart and fin tip (Yin, Lepilina, Smith, & Poss, 2012; Yin et al., 2008). miR-133 restricts cardiomyocyte proliferation during heart regeneration, acting by targeting cell junction components *cln5* and *cx43* and cell cycle factors *mps1*, *PA2G4*, and *cdc37* (Yin et al., 2012) under control of fibroblast growth factor signaling and *mps1* during appendage regeneration (Yin et al., 2008). Of note, the miR-133-*mps1* regulatory unit participates in two different regenerative contexts, heart and fin, making this unit of high interest for generic tissue regeneration (Yin et al., 2008, 2012). Conversely, a member of the same family, miR-133b, is upregulated and is an important factor during zebrafish spinal cord regeneration through its interaction with RhoA mRNA, resulting in reduction of RhoA protein levels (Y. M. Yu et al., 2011). This finding supports the hypothesis that the same miRNA may have multiple functions and members of the same miRNA family may have opposite functions during regeneration.

3 | AXOLOTL TAIL REGENERATION

The regeneration ability of salamanders, including the axolotl, involves changes in the expression level of 78 highly conserved miRNAs during early tail blastema formation and spinal cord regeneration (Sehm et al., 2009). Of these, miR-196 was found to be an essential regulator, primarily upregulated in cells within 500 μ m of the amputation plane. Direct inhibition of the expression of miR-196 via injection and electroporation of miRIDIAN miRNA inhibitors into the cells of the blastema was noted at the time of amputation and 2 days postamputation. The inhibition of miR-196 caused defects in regeneration, decreasing cell proliferation in the blastema without causing apoptosis. Moreover, axolotls treated with inhibitor-196 showed upregulation of *Pax7*, bone morphogenetic protein 4 (*BMP4*), and *Msx1* genes during tail regeneration, indicating a possible role of miR-196 in modulating these genes, especially because *Pax7* displays

a seven mer binding site for miR-196 at the 3'-UTR (Sehm et al., 2009). Similarly, miR-206 regulates *Pax7* expression via binding the 3'-UTR, resulting in downregulation of *Pax7*. This downregulation induces skeletal muscle satellite cell and myoblast differentiation (Chen et al., 2010; Dey, Gagan, & Dutta, 2011). miR-206 is downregulated 3 days after injury in the axolotl. By the 7th day, miR-206 starts to show upregulation (Gearhart, Erickson, Walsh, & Echeverri, 2015). Repressing *Pax7* acts as a signal to the cell to proliferate and migrate to form a new ependymal tube (Buckingham & Relaix, 2007; Sehm et al., 2009). However, *Pax7* is expressed in proliferating myoblasts and suppresses differentiation (Dey et al., 2011). Thus, *Pax7* downregulation at a specific stage is essential for completion of the regeneration process. Unlike the axolotl, which lacks *Pax3* (Nowoshilow et al., 2018), the Iberian ribbed newt *Pleurodeles waltl* requires *Pax3* for regeneration, even without functional *Pax7* gene (Elewa et al., 2017).

Amex-miR-pn2533, -pn3918, and -pn2611 are significantly upregulated and enriched in the early tail blastema but are not expressed in mature uninjured tissues. *Amex*-miR-pn2533 and -pn3918 return to basal levels by 7 days after injury. In this context, these miRNAs may be coexpressed with other miRNAs and could help maintain a population of undifferentiated cells. *Amex*-miR-pn3918 is expressed during the regeneration of other tissues, including skin cells and neural stem cells. In addition, this miRNA plays a role in the differentiation of the tail. When inhibited, defects in the tail structures with fewer axons reinnervating the tail are observed 3–4 days after injury (Gearhart et al., 2015). Similarly, miR-21 is upregulated throughout the course of regeneration of axolotl tail and limb and reaches peak expression ~7 days after injury (Gearhart et al., 2015; Holman, Campbell, Hines, & Crews, 2012). miR-21 regulates levels of Jagged-1-a, a Notch ligand, which is responsible for signals affecting cell growth and differentiation during tissue regeneration. This action occurs through SRY-related HMG box transcription factor 9 (*Sox9*) seen in liver regeneration (Köhler et al., 2004; Russell, Ko, Monga, & Shin, 2019; Vieira et al., 2015).

Interestingly, downregulation of let-7 family members is a primary contributor to early-stage reprogramming or induction of pluripotency (Figure 2) in various tissues or cells, including fibroblasts, stem cells, and oocytes (Nakamura et al., 2010; Tsonis et al., 2007; Unternaehrer et al., 2014; Worringer et al., 2014), possibly via inhibitory binding of epithelial-to-mesenchymal transition factor *SNAIL1* (*SNAIL*) to the let-7 promoter (Unternaehrer et al., 2014). In zebrafish, let-7a/c and miR-99/100 are critical regulators of heart regeneration, and their downregulation triggers dedifferentiation of cardiomyocytes (Aguirre et al., 2014). let-7i is responsible for regulating inflammation to levels suitable for zebrafish heart regeneration via tumor necrosis factor (TNF). Continued upregulation of let-7i is essential for heart regeneration, possibly by maintaining cardiomyocyte proliferation (A. M. Smith, Dykeman, King, & Yin, 2019). let-7 family members are downregulated or upregulated depending on the stage of heart regeneration (A. M. Smith et al., 2019), but let-7a is significantly upregulated in early tail blastema until 14 days after injury. At this time, cells begin to differentiate to replace lost tissues (Gearhart et al., 2015). let-7a directs key

regenerative and developmental events that involved in limb and neurons formation (Rybak et al., 2008; Schulman, Esquela-Kerscher, & Slack, 2005; Wulczyn et al., 2007). This information may support the hypothesis that reprogramming of cells is an aspect of regeneration in the axolotl or possibly that other molecules are responsible for the reprogramming.

Based on these findings, the first 14 days are critical for the control of regeneration, in which a starting point for the process is established in the axolotl. This period may be a feature specific to tail regeneration.

4 | AXOLOTL LIMB REGENERATION

At a cellular level, limb regeneration in the axolotl has been well studied. Generally, three main events occur: wound healing, subsequent blastema formation, and finally limb redevelopment. After injury, neighboring epithelial cells rapidly cover exposed areas, forming a wound epidermis (Campbell & Crews, 2008). This process induces signals for the formation of blastema by a heterogeneous mass of mesenchymal cells. These cells change in morphology, proliferate, and proceed to form a new appendage (Brockes & Kumar, 2002; Kragl et al., 2009). Many genes expressed in developing limbs can be re-expressed during regeneration (Haas & Whited, 2017). This process is poorly understood at a molecular level (Holman et al., 2012). Gene expression during axolotl limb regeneration diverges progressively through time with punctuated episodes of transcription that coincides with stages of limb regeneration: amputation, early bud, late bud, and pallet (Voss et al., 2015). According to Monaghan et al. (2012), many of these genes are activated before blastema outgrowth and expressed in specific tissue types that include epidermis, peripheral nerve, and mesenchyme.

Recently, certain miRNAs were shown to play an important role in regulation of tissue-specific gene development (Guo et al., 2014; King & Yin, 2016; Sood, Krek, Zavolan, Macino, & Rajewsky, 2006). Therefore, miRNAs may also regulate characteristic rapid changes in gene expression during limb injury-induced regeneration. This regulation was identified in newt lens regeneration (Makarev, Spence, Del Rio-Tsonis, & Tsonis, 2006; Nakamura et al., 2010; Tsonis et al., 2007). However, miRNA involvement in salamander limb regeneration is poorly understood. Y. Yu et al. (2019) recently concluded that every stage of regeneration is dynamically regulated by specific miRNAs. For example, miR-737 showed lower levels of expression during wound healing, 21 miRNAs were upregulated and 15 miRNAs were downregulated during blastema formation, and 9 were upregulated and 31 were downregulated during redevelopment in *C. orientalis*. Differentially expressed miRNAs such as miR-21 and miR-142 were detected throughout the regeneration period. miR-223 specifically regulated expression of the collagen alpha-3 (IX) chain (COL9A3) during wound healing and blastema formation, and miR-133a regulated the expression of glucose-6-phosphate dehydrogenase (G6PD) and endophilin-A2 (SH3G1) during blastema formation and redevelopment (Y. Yu et al., 2019). King and Yin (2016) reported a set of miRNAs that showed differential expression during

blastema formation in zebrafish caudal fin, bichir pectoral fin, and axolotl forelimb. Across all three regenerating systems, miR-21 was the most highly expressed and upregulated miRNA, having a modulating effect on at least 19 blastema-associated gene transcripts. Holman et al. (2012) also reported overexpression of miR-21 in mid-bud blastema during axolotl limb regeneration, and *Jagged1* was a putative target developmental gene. miR-21 is, thus, a strong candidate for a key functional role in limb regeneration. Presumably, it maintains blastema cell differentiation via negative regulation of *Jagged1*. Brito (2018) indicated that miR-10b-5p might be implicated in the limb regenerative response of the newt. Downregulation of this miRNA during limb regeneration was reported, which apparently is necessary to ensure appropriate regulation of the regeneration process. An overexpression of miR-10b-5p can delay regeneration by affecting blastema formation. Interestingly, miR-427, a miRNA that rapidly clears cache of maternally supplied mRNAs by deadenylation after the midblastula transition in *Xenopus laevis* embryo (Lund, Liu, Hartley, Sheets, & Dahlberg, 2009), seemed to have a similar function during *P. waltl* limb regeneration. Thereby, miR-427 clears a cell's "working memory" without affecting the cell's identity and functional "deep memory" and thus allows dedifferentiation of cells during the regeneration (Elewa et al., 2017).

Many miRNAs are highly conserved and are often identified as important regulators across multiple model regeneration systems. Their study allows the identification of miRNA families and putative novel miRNAs associated with blastema formation. Expanding the knowledge of key regulatory roles for miRNAs during the natural process of cellular reprogramming networks required for limb regeneration is critical to understanding regenerative systems (Gearhart et al., 2015; Holman et al., 2012).

5 | HEDGEHOG SIGNALING, miRNA, AND REGENERATION

miRNA is known to regulate multiple genes at the posttranscriptional level, subsequently modulating a variety of signaling pathways, including hedgehog signaling. Hedgehog signaling has been shown to critically regulate aspects of development and regeneration, for example, positional identity and cellular maturation (Singh et al., 2015). Hedgehog signaling is required for tail and limb regeneration in species with substantial regenerative capacity, including the axolotl (Schnapp, Kragl, Rubin, & Tanaka, 2005; Singh et al., 2015; Torok, Gardiner, Izpisua-Belmonte, & Bryant, 1999). Sonic hedgehog signaling is required for Müller glia dedifferentiation during retina regeneration in zebrafish (Kaur et al., 2018). Hedgehog signaling in regeneration has been reviewed by Singh et al. (2015). Localized expression of sonic hedgehog was detected in the ventral floor plate domain in mature and regenerating axolotl spinal cord. The expression of *Patched1* receptor indicates that hedgehog signaling also occurs in axolotl blastema. It is required for patterning blastema tissue and muscle progenitors, as evidenced by expression analysis of *Sox9* and *Pax7*. In addition, hedgehog signaling controls blastema cell

proliferation (Schnapp et al., 2005). miRNA-mediated hedgehog signaling in axolotl regeneration has not been studied.

Hedgehog signaling regulates tissue reconstruction in damaged liver (Hyun, Wang, Kim, Kim, & Jung, 2015). miR-125b attenuates expression of hedgehog signaling, primarily targeting the Smoothed (Smo) gene. This gene promotes reduced liver fibrosis (Hyun et al., 2015). Inhibition of hedgehog signaling during somite and neuronal development in zebrafish embryos results in considerable changes in the expression of several miRNA. Of these, miR-214 was downregulated (Thatcher, Flynt, Li, Patton, & Patton, 2007). miR-214 enhances the response to different levels of hedgehog signals. Repressing miR-214 function causes increased expression of suppressor of fused (*su(fu)*), which decreases *Patched1* expression in adaxial cells and increases expression in lateral presomitic mesoderm. These modulations alter hedgehog-mediated cell fate specification during somatogenesis, particularly in muscle cells (Flynt, Li, Thatcher, Solnica-Krezel, & Patton, 2007). Kaur et al. (2018) suggested that lin28a-mediated suppression of let-7 is essential for translational regulation of sonic hedgehog signaling in Müller glia-derived progenitor cells.

6 | RETINOIC ACID INDUCES GENE EXPRESSION IN REGENERATION

The active vitamin A metabolite retinoic acid (RA) has a vital function during vertebrate development (Duester, 2008; Marlétaz, Holland, Laudet, & Schubert, 2006). This lipophilic molecule acts in the nucleus and induces gene transcription via ligand-activated nuclear transcription factors, which include RA (RAR α , RAR β , and RAR γ) and retinoid X (RXR α , RXR β , and RXR γ) receptors (Mark, Ghyselinck, & Chambon, 2006). RARs heterodimerize to RXRs, forming a transcriptional complex (RAR/RXR), and bind to retinoic acid DNA response elements (RAREs) located near RA target genes (Chambon, 1996). *Hox* genes regulate the signaling functions of RA, which may be a key to patterning in all embryonic tissue layers (Garcia-Fernández, 2005). RA is also involved in regeneration, causing "super-regeneration" in species possess unique regenerative abilities in which RA induces regeneration of extra tissue. This activity of RA has even been reported in mammalian lung alveoli (Maden, 1982; Maden & Hind, 2003; Maden & Hind, 2004). The key active function of RA during limb regeneration is highlighted by its effect on re-specifying the proximo-distal axis of regenerating limbs (Monaghan & Maden, 2012; Scadding & Maden, 1994). RA signaling seems to be most important in the apical epithelium during limb regeneration. During development, its presence has been reported in the developing forelimb bud mesenchyme, suggesting a different role of RA during limb regeneration (Monaghan & Maden, 2012). Disruption in RA signaling can lead to detrimental effects in both limb development and regeneration (Monaghan & Maden, 2012). To illustrate, a limited concentration of RA can inhibit RAR signaling or RA metabolism (Nguyen et al., 2017). In excess concentration, RA can induce duplicate patterning during limb regeneration, as reported in *Xenopus* and

salamanders (Cuervo & Chimal-Monroy, 2013; Kim & Stocum, 1986; Nguyen et al., 2017). Regeneration has been suggested as a vestige of development because both *Xenopus* and salamanders use the same gene pathways (Maden & Hind, 2003). Nguyen et al. (2017) reported that differential function of RA between developing and regenerating limbs might be due to interactions with RARs during specific stages of regeneration or in specific cell types.

More than 532 genes have been identified as regulatory targets of RA; 27 are driven directly by a ligand heterodimer RAR/RXR receptor to a RARE, whereas the remainder may be regulated indirectly by RA thought to be intermediate transcription factors (Balmer & Blomhoff, 2002). Genes associated with regulation of RA signaling during limb regeneration process include *aldh1a3*, *crabp1*, *crabp2*, *rdh10*, *Raldh3*, and *Raldh1* (Blum & Begemann, 2012; Knapp et al., 2013; McEwan, Lynch, & Beck, 2011; Monaghan & Maden, 2012). Monaghan and Maden (2012) reported exclusive expression of *crabp1* in the limb mesenchyme of innervated limbs and *crabp2* upregulation in innervated, denervated limbs, and flank wounds. Proteins produced by these two genes (cellular retinoic acid binding proteins, CRABP) are associated with control of nuclear uptake of RA, which regulates transactivation of nuclear RARs (Donovan, Olofsson, Gustafson, Dencker, & Eriksson, 1995). Inhibition of *Raldh* can block axolotl limb regeneration (Scadding, 1999). RA also mediates early events in limb regeneration by regulating expression of molecules involved in the formation of wound epidermis and initiation of dedifferentiation through upregulation of matrix metalloproteinases (MMPs; Lee, Ju, & Kim, 2012). Identification of such genes is important to understanding cellular and molecular mechanisms that drive regeneration and can be a useful resource for analyzing the specificity of genes at each stage of the regeneration process.

Limb regeneration can be also regulated at the posttranscriptional level by miRNAs (King & Yin, 2016). However, few studies have identified a role for miRNAs during RA signaling in regeneration-competent species. In mollusks, miRNAs were shown to be important regulators of RA-induced neuronal outgrowth and regeneration (Walker, Spencer, Necakov, & Carlone, 2018). In addition, A. C. Lepp and Carlone (2014) has provided information that miR-133a can act as an upstream regulator of RAR β 2 expression, whereas miR-1 can cause a significant decrease in RAR β 2 during caudal spinal cord regeneration in the newts. Both miRNAs may participate with RAR β 2 in a negative feedback loop (Amanda C. Lepp & Carlone, 2015), a possible role for regulation of RA signaling. The study of miRNAs and target genes can serve as a platform to inform new hypotheses about miRNA regulation of limb/appendage regeneration.

7 | POTENTIAL CONTRIBUTIONS OF THE IMMUNE SYSTEM IN REGENERATION IN THE CONTEXT OF miRNA

The immune system plays an essential and effective role in the regeneration process. There is a strong link between antioxidant

defense machinery and immune systems (Abdel-Mageid, Zaki, El Senosi, El Asely et al., 2020; Abdel-Mageid, Zaki, El Senosi, Fahmy et al., 2020; Abo-Al-El, 2018a, 2018b, 2019a). Signaling of reactive oxygen species (ROS) after axolotl tail amputation is required for early phases of regeneration. Confirmation for this mechanism involved showing reduced cellular proliferation and inhibition of tail outgrowth after treatment with ROS producing enzymes inhibitors (Al Haj Baddar, Chithrala, & Voss, 2019). ROS are important to achieve optimal T-cell receptor signaling. miRNAs play a modulator role in this respect. For example, miR-23a suppresses ROS, enhances CD4⁺ T-cell survival, and thereby curbs necrosis (Zhang et al., 2016). These results are consistent with interlaced pathways among miRNAs, ROS, and the immune system in regeneration.

Further, systemic macrophage depletion during the first 24 hr after limb amputation in the axolotl resulted in wound closure with extensive fibrosis, and permanent impairment of limb regeneration, accompanied by dysregulation of specific gene pathways essential for progression from wound healing to regeneration (Godwin, Pinto, & Rosenthal, 2013). miRNAs are important for the maturation of macrophages via toll-like receptors, and they enhance macrophage function after stimulation (Lu & Liston, 2009). This finding is consistent with let-7i expression primarily localized to macrophages and epicardial cells, where it coordinates epicardium migration and cardiomyocyte proliferation in concert with the TNF signaling pathway in zebrafish. let-7i adjusts the activity of TNF to levels for a "pro-regenerative microenvironment" (A. M. Smith et al., 2019). Nuclear factor kappa light-chain enhancer of activated B cell (NF- κ B) signaling is stimulated early during liver regeneration, in which upregulation of miR-21 results from direct targeting of *Pellino-1* (Marquez, Wendlandt, Galle, Keck, & McCaffrey, 2010). miR-21 has anti-inflammatory and neuroprotective functions and improves functional recovery. In addition, upregulated miR-21 reduces astrocyte hypertrophy after spinal cord injury. This activity is due to interactions with other immune-related molecules, such as toll-like receptor 4-myeloid differentiation primary response 88-NF κ B signaling, interleukin-6 (IL-6), IL-10, and signaling pathways of host signal transducer and activator of transcription (stat) 3 (Gaudet, Fonken, Watkins, Nelson, & Popovich, 2018).

8 | STEM CELL AND ROS INTERACTIONS WITH miRNA DURING REGENERATION

Interestingly, miRNAs shows cross-signal with ROS during cell proliferation and regeneration. In the axolotl, stem cells may contribute to the formation of blastema (Kragl et al., 2009; Sobkow, Epperlein, Herklotz, Straube, & Tanaka, 2006). ROS regulates miR-210 expression that, in turn, regulates migration and proliferation of adipose-derived stem cells. miR-210 increases ROS generation via downregulation of iron-sulfur cluster scaffold homolog 2 (Kim, Park, Song, Kim, & Sung, 2013). In the green anole lizard, a set of miRNAs, including miR-133a, miR-133b, and miR-206, have been implicated in regulation of regeneration and stem cell proliferation (Hutchins

et al., 2016). Thus, stem cells could have a role in the regeneration process. However, dedifferentiation and transdifferentiation of blastema cells may also be a source of cells for a regenerated organ (Morrison, Löff, He, & Simon, 2006; Poss, 2010). For example, skeletal muscle dedifferentiation involves satellite cell (muscle stem cell) activation and contributes to limb regeneration in salamanders (Morrison et al., 2006).

9 | CONCLUSION AND FURTHER PERSPECTIVES

A diverse landscape of transcripts per tissue, including protein-coding RNA and miRNA has been reported in axolotl. Of these, 297 novel putative mature miRNAs and 352 conserved miRNAs were identified. Approximately, 176 novel miRNAs expressed during development are normally expressed in the axolotl tissues, along with 28 miRNAs target transcripts involved in controlling neuron development. Expression of miR-122, miR-233, miR425-5p, and miR3600 was high in regenerating liver, and expression of miR-429 was low (Caballero-Pérez et al., 2018), indicting the crucial role of these miRNAs. Moreover, principal component analysis for transcript expression in the absence of regeneration in many axolotl tissues was similar to that for transcripts found in limbs and tail. Transcripts in heart and liver were more diverse, reflecting the specialized function of such organs. Gene Ontology enrichment analysis showed that heart and liver-enriched genes in the axolotl are also specific to liver and heart in humans, including transcripts associated with disorders and congenital diseases. The axolotl is, thus, a candidate organism for studying heart and liver diseases that might be, in the future, addressed through regenerative medicine (Caballero-Pérez et al., 2018).

The axolotl has unique regenerative abilities, and miRNAs are intimately involved in its regulation. Regeneration is a complex process influenced by many factors. miRNAs regulate gene expression in several metabolic pathways implicated in regeneration. miRNAs act to manipulate the timely expression of regulatory proteins to coordinate the various processes necessary to reform new organs and tissues. miRNAs, thus, apparently organize when, which, and how cells participate in the regeneration process. More focused studies are needed to identify a role for miRNA in initiating axolotl regeneration, which presumably could be used with other organisms that lack the exceptional regenerative capabilities of this species. ncRNAs, including miRNA, are cornerstones for regeneration, and comparative analysis among organisms with high or low regenerative capacity could be illuminating (King et al., 2018). miRNAs have potential advantages; for example, mature miRNAs could be obtained by chemical synthesis, thus obtaining more precise sequences with very low incidence of mutations (Wang, Li, Gao, Sun, & Fang, 2015). In addition, lipid-based carriers of miRNA could be successfully developed (Vickers & Remaley, 2012) or possibly modified micro-organisms could be used to effectively deliver and express desired miRNA(s) within target (Abo-Al-El, 2020). miRNAs might become

useful therapeutic molecules and be easily manipulated in regeneration and other biological studies.

CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

AUTHOR CONTRIBUTIONS

B. A. (primary) and H. G. A. A. E. (shared) wrote discussions on limb regeneration. B. A. wrote the discussion on retinoic acid and regeneration. A. A. E. wrote the remainder of the manuscript and designed and prepared the figures. Both authors read and approved the manuscript.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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